

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication	20250273378
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	SYU; Yong-Long et al.

FILTER STRUCTURE

Abstract

A filter structure is provided. The filter structure includes a substrate, a capacitor element, and a first choke element. The capacitor element and the first choke element are disposed on the substrate. The first choke element has a first ferrite core, two first windings wound around the first ferrite core, and a first magnetic conductive member. The first ferrite core is received in the first magnetic conductive member and located adjacent to the first choke element.

Inventors: SYU; Yong-Long (TAIPEI CITY, TW), CHEN; Kai-De (TAIPEI CITY, TW), CHEN; Chen (TAIPEI CITY, TW)

Applicant: LITE-ON TECHNOLOGY CORPORATION (TAIPEI CITY, TW)

Family ID: 1000008170190

Appl. No.: 18/830655

Filed: September 11, 2024

Foreign Application Priority Data

TW	113131268	Aug. 20, 2024
----	-----------	---------------

Related U.S. Application Data

us-provisional-application US 63557626 20240226

Publication Classification

Int. Cl.: H03H7/01 (20060101)

U.S. Cl.:

CPC H01F27/06 (20130101); H01F27/266 (20130101); H01G2/06 (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/557,626, filed on Feb. 26, 2024, the entirety of which is incorporated by reference herein. This application claims priority of Taiwan Patent Application No. 113131268, filed on Aug. 20, 2024, the entirety of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to filter structure, and, in particular, to a filter structure that has at least a magnetic conductive member.

Description of the Related Art

[0003] Traditional EMC filters can easily be affected by the limitations of their compact size and the mutual coupling effect between inductors and capacitors, which leads to problems such as a decrease in the filter's ability to suppress electromagnetic noise.

[0004] Currently, it is a common practice to increase the number of windings in a common-mode choke in the filter, and to suppress electromagnetic noise by increasing the inductance. However, this method may increase the DC resistance loss of the EMC filter and cause the filter's performance to degrade.

[0005] In view of this, it has become a challenge to design a filter structure that can have both a small size and high performance.

BRIEF SUMMARY OF THE INVENTION

[0006] To address the aforementioned problems, an embodiment of the present invention provides a filter structure that includes a substrate, a capacitor element, and a first choke element. The capacitor element and the first choke element are disposed on the substrate. The first choke element includes a first ferrite core, two first windings, and a first magnetic conductive member, and the first windings are wound around the first ferrite core. The first magnetic conductive member surrounds the first ferrite core and is located adjacent to the capacitor element.

[0007] In some embodiments, the filter structure further includes a second choke element disposed on the substrate, wherein the capacitor element is situated between the first choke element and the second choke element. The second choke element has a second ferrite core, two second windings and a second magnetic conductive member, and the second windings are wound around the second ferrite core, wherein the second magnetic conductive member surrounds the second ferrite core and is located adjacent to the capacitor element.

[0008] In some embodiments, a part of the first magnetic conductive member is located between the first windings.

[0009] In some embodiments, the substrate forms a plurality of first slots, and the first magnetic conductive member extends through the first slots.

[0010] In some embodiments, the first magnetic conductive member has a rectangular hollow structure.

[0011] An embodiment of the invention further provides a filter structure that includes a substrate, a capacitor element, and a first choke element. The capacitor element and the first choke element

are disposed on the substrate. The first choke element includes a first ferrite core, two first windings and a first magnetic conductive member, the first windings are wound around the first ferrite core, and the first ferrite core is situated in the first magnetic conductive member. The first magnetic conductive member is located adjacent to the capacitor element and forms a first opening.

[0012] In some embodiments, the first magnetic conductive member has a U-shaped structure, and the first opening and the substrate are located on opposite sides of the first ferrite core.

[0013] In some embodiments, the first magnetic conductive member has an inverted U-shaped structure, and the first opening faces the substrate.

[0014] In some embodiments, the filter structure further includes a second choke element disposed on the substrate, wherein the capacitor element is situated between the first and second choke elements, and the second choke element has a second ferrite core, two second windings and a second magnetic conductive member. The second windings are wound around the second ferrite core, the second ferrite core is situated in the second magnetic conductive member, and the second magnetic conductive member is located adjacent to the capacitor element and forms a second opening.

[0015] In some embodiments, the first magnetic conductive member and the second magnetic conductive member both have a C-shaped structure, and the first and second openings face opposite directions.

[0016] An embodiment of the invention further provides a filter structure that includes a substrate, a capacitor element, and a first choke element. The capacitor element and the first choke element are disposed on the substrate and located adjacent to the capacitor element. The first choke element includes an annular first ferrite core, two first windings and a first magnetic conductive member, the first ferrite core forms a first through hole, the first windings are wound around the first ferrite core, and the first magnetic conductive member is disposed on the first ferrite core. At least a part of the first through hole is covered by the first magnetic conductive member.

[0017] In some embodiments, the projections of the capacitive element and the first ferrite core on a plane perpendicular to the central axis of the first through hole at least partially overlap.

[0018] In some embodiments, the first magnetic conductive member contacts the first ferrite core, wherein a gap is formed between the first ferrite core and each of the first windings.

[0019] In some embodiments, the first magnetic conductive member has a first section situated between the first ferrite core and the capacitive element.

[0020] In some embodiments, the first magnetic conductive member further has a second section connected to and bent relative to the first section.

[0021] In some embodiments, the first magnetic conductive member further has a third section disposed on the substrate, and the first section is connected between the second and third sections.

[0022] In some embodiments, the filter structure further includes a second choke element disposed on the substrate, wherein the second choke element has an annular second ferrite core, two second windings and a second magnetic conductive member. The second ferrite core forms a second through hole, the second windings are wound around the second ferrite core, and the second magnetic conductive member is disposed on the second ferrite core. At least a part of the second through hole is covered by the second magnetic conductive member.

[0023] In some embodiments, the first choke element, the capacitor element, and the second choke element are sequentially arranged along the central axis of the first through hole.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The present invention can be more fully understood by reading the subsequent detailed description and examples with references made to the accompanying drawings, wherein:

[0025] FIG. **1** shows a perspective diagram of a filter structure **100** in accordance with an embodiment of the invention.

[0026] FIG. **2** shows another perspective diagram of the filter structure **100** in FIG. **1**.

[0027] FIG. **3** shows an exploded view of the filter structure **100** in FIGS. **1** and **2**.

[0028] FIG. **4** shows an exploded view of a filter structure **200** in accordance with another embodiment of the invention.

[0029] FIG. **5** is a perspective diagram of the filter structure **200** in FIG. **4**.

[0030] FIG. **6** is another perspective diagram of the filter structure **200** in FIG. **5**.

[0031] FIG. **7** shows an exploded view of a filter structure **300** in accordance with another embodiment of the invention.

[0032] FIG. **8** shows a perspective view of a filter structure **400** in accordance with another embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

[0033] The making and using of the embodiments of the filter structure are discussed in detail below. It should be appreciated, however, that the embodiments provide many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use the embodiments, and do not limit the scope of the disclosure.

[0034] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. It should be appreciated that each term, which is defined in a commonly used dictionary, should be interpreted as having a meaning conforming to the relative skills and the background or the context of the present disclosure, and should not be interpreted in an idealized or overly formal manner unless defined otherwise.

[0035] In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings, and in which specific embodiments of which the invention may be practiced are shown by way of illustration. In this regard, directional terminology, such as “top,” “bottom,” “left,” “right,” “front,” “back,” etc., is used with reference to the orientation of the figures being described. The components of the present invention can be positioned in a number of different orientations. As such, the directional terminology is used for the purposes of illustration and is in no way limiting.

[0036] FIG. **1** shows a perspective diagram of a filter structure **100** in accordance with an embodiment of the invention. FIG. **2** shows another perspective diagram of the filter structure **100** in FIG. **1**. FIG. **3** shows an exploded view of the filter structure **100** in FIGS. **1** and **2**.

[0037] Referring to FIGS. **1-3**, the filter structure **100** may comprise an EMC filter that can be disposed in a circuit system of a charging pile, energy storage equipment, computer, server, or adapter.

[0038] In this embodiment, the filter structure **100** primarily includes a substrate B, a first choke element **11**, a second choke element **12**, and a capacitor element C. The substrate B may comprise a printed circuit board (PCB). Additionally, the first choke element **11**, the second choke element **12**, and the capacitor element C are disposed on the substrate B, wherein the first and second choke elements **11** and **12** are used as a common-mode chock filter.

[0039] It should be noted that the capacitor element C is positioned between the first choke element **11** and the second choke element **12**. The first choke element **11**, the second choke element **12**, and the capacitor element C are electrically connected to each other via the conductors of the substrate B to form an EMC filter circuit.

[0040] As shown in FIGS. **1-3**, the first choke element **11** primarily includes an annular first ferrite core E**11**, two symmetrical first windings W**11** and a first magnetic conductive member S**11**, wherein the first windings W**11** are wound around the first ferrite core E**11** and electrically connected to the substrate B. Moreover, the first magnetic conductive member S**11** has a

rectangular hollow structure and surrounds the first ferrite core **E11**. In particular, a part of the first magnetic conductive member **S11** is situated between the two first windings **W11**, and the capacitive element **C** is located adjacent to the first magnetic conductive member **S11**.

[0041] In this embodiment, the first ferrite core **E11** has a first through hole **E110**. The first magnetic conductive member **S11** is disposed on the first ferrite core **E11** and covers at least a part of the first through hole **E110**.

[0042] Similarly, the second choke element **12** primarily includes an annular second ferrite core **E12**, two symmetrical second windings **W12** and a second magnetic conductive member **S12**, wherein the second windings **W12** are wound around the second ferrite core **E12** and electrically connected to the substrate **B**. The second magnetic conductive member **S12** has a rectangular hollow structure and surrounds the second ferrite core **E12**. A part of the second magnetic conductive member **S12** is situated between the two second windings **W12**, and the capacitive element **C** is located adjacent to the second magnetic conductive member **S12**.

[0043] In this embodiment, the second ferrite core **E12** has a second through hole **E120**. The second magnetic conductive member **S12** is disposed on the second ferrite core **E12** and covers at least a part of the second through hole **E120**.

[0044] The substrate **B** is located below the first and second ferrite cores **E11** and **E12**. Two elongated first slots **B1** and two elongated second slots **B2** are formed on the substrate **B**. Specifically, the first magnetic conductive member **S11** extends through the first slots **B1**, and the second magnetic conductive member **S12** extends through the second slots **B2**, whereby the first magnetic conductive member **S11** and the second magnetic conductive member **S12** are fixed to the substrate **B**. Here, the capacitive element **C** is located between the first magnetic conductive member **S11** and the second magnetic conductive member **S12** along the **X** axis.

[0045] For example, the first magnetic conductive member **S11** and the second magnetic conductive member **S12** are both magnetic conductive plates with high magnetic permeability. The first and second magnetic conductive members **S11** and **S12** may comprise nanocrystalline material, but they are not limited to those disclosed in each embodiment of the invention.

[0046] With the first magnetic conductive member **S11** and the second magnetic conductive member **S12** respectively disposed in the first choke element **11** and the second choke element **12**, coupling noise effect between the windings or between the windings and the capacitor can be significantly reduced. Moreover, the leakage inductance of the common-mode choke circuit can be greatly increased, thereby reducing the loss of the EMC filter and improving the performance of the filter structure **100**.

[0047] In particular, the first choke element **11**, the capacitor element **C** and the second choke element **12** in this embodiment are sequentially arranged along the central axis **A1** (**X** direction) of the first through hole **E110** and the second through hole **E120**. It should be noted that the projections of the capacitive element **C**, the first ferrite core **E11**, and the second ferrite core **E12** on a plane perpendicular to the central axis **A1** (**X** direction) at least partially overlap. Therefore, the performance of the filter structure **100** can be improved, and space utilization of the filter structure **100** can also be increased to achieve miniaturization of the product.

[0048] In some embodiments, the top portion of the first magnetic conductive member **S11** may contact the first ferrite core **E11**, and a gap is formed between the first magnetic conductive member **S11** and each of the two first windings **W11**. Namely, the first magnetic conductive member **S11** contacts the first ferrite core **E11**, but does not contact the first windings **W11**. Similarly, the top portion of the second magnetic conductive member **S12** may contact the second ferrite core **E12**, and a gap is formed between the magnetic component **S12** and each of the two second windings **W12**. Namely, the second magnetic conductive component **S12** contacts the second ferrite core **E12**, but does not contact the second winding **W12**.

[0049] In some embodiments, the second choke element **12** may be omitted from the filter structure **100**, and only the first choke element **11** and the capacitor element **C** are disposed on the substrate

B. However, the configuration of the filter structure **100** is not limited to those disclosed in the embodiments of the invention.

[0050] FIG. **4** shows an exploded view of a filter structure **200** in accordance with another embodiment of the invention. Additionally, FIG. **5** is a perspective diagram of the filter structure **200** in FIG. **4**, and FIG. **6** is another perspective diagram of the filter structure **200** in FIG. **5**.

[0051] As shown in FIGS. **4**, **5**, and **6**, the filter structure **200** is different from the filter structure **100** of FIGS. **1** to **3** in that the first magnetic conductive element **S21** of the first choke element **21** and the second magnetic conductive element **S22** of the second choke element **22** both have a U-shaped structure. Here, the first magnetic conductive element **S21** forms a first opening **S210** facing the Z direction, and the second magnetic conductive member **S22** forms a second opening **S220** facing the Z direction. That is, the first opening **S210** and the second opening **S220** face the same direction.

[0052] In this embodiment, the first choke element **21**, the capacitor element **C** and the second choke element **22** are sequentially arranged along the central axis **A2** of the first and second through holes **E210** and **E220**. The annular first ferrite core **E21** is received in the U-shaped first magnetic conductive member **S21**, the annular second ferrite core **E22** is received in the U-shaped second magnetic conductive member **S22**, and the capacitive element **C** is disposed between the first conductive member **S21** and the second magnetic conductive member **S22**.

[0053] In addition, the first opening **S210** of the first magnetic conductive member **S21** is located adjacent to the top side of the first ferrite core **E21**, the second opening **S220** of the second magnetic conductive member **S22** is located adjacent to the top side of the second ferrite core **E22**, and the substrate **B** is situated on the bottom side of the first and second ferrite cores **E11** and **E22**. That is, the first opening **S210**, the second opening **S220**, and the substrate **B** are located on opposite sides of the first and second ferrite cores **E21** and **E22**.

[0054] With the first magnetic conductive member **S21** and the second magnetic conductive member **S22** respectively disposed in the first choke element **21** and the second choke element **22**, coupling noise effect between the capacitor element **C** and the first and the second windings **W21** and **W22** can be significantly reduced, whereby the performance of the filter structure **100** can be improved. Moreover, with the first opening **S210** and the second opening **S220** respectively formed on the top side of the first and second magnetic conductive members **S21** and **S22**, the size and height of the filter structure **200** along the Z axis can be effectively reduced, thereby saving costs and achieving miniaturization of the product.

[0055] In some embodiments, the second choke element **22** may be omitted from the filter structure **200**, and only the first choke element **21** and the capacitor element **C** are disposed on the substrate **B**. However, the configuration of the filter structure **200** is not limited to those disclosed in the embodiments of the invention.

[0056] FIG. **7** shows an exploded view of a filter structure **300** in accordance with another embodiment of the invention. As shown in FIG. **7**, the filter structure **300** is different from the filter structure **200** of FIGS. **4** to **6** in that the first magnetic conductive element **S31** of the first choke element **31** and the second magnetic conductive element **S32** of the second choke element **32** both have an inverted U-shaped structure. Here, the first magnetic conductive element **S31** forms a first opening **S310** facing the -Z direction, and the second magnetic conductive member **S32** forms a second opening **S320** facing the -Z direction. That is, the first opening **S310** and the second opening **S320** face the same direction.

[0057] In this embodiment, the first choke element **31**, the capacitance element **C** and the second choke element **32** are sequentially arranged along the central axis **A3** of the first and second through holes **E310** and **E320**. The annular first ferrite core **E31** is received in the first magnetic conductive member **S31**, and the annular second ferrite core **E32** is received in the second magnetic conductive member **S32**. Both ends of the first magnetic conductive member **S31** can be engaged in the first slots **B1** of the substrate **B** after assembly, and both ends of the second magnetic

conductive member **S32** can be engaged in the second slots **B2** of the substrate **B** after assembly. [0058] Additionally, it can be seen in FIG. 7 that the first opening **S310** of the first magnetic conductive member **S31** faces the substrate **B**, the second opening **S320** of the second magnetic conductive member **S32** also faces the substrate **B**, and the capacitive element **C** is located between the first magnetic conductive part **S31** and the second magnetic conductive part **S32** along the **X** axis.

[0059] In this embodiment, with the inverted U-shaped first and second magnetic conductive members **S31** and **S32**, coupling noise effect between the capacitor element **C** and the first and the second windings **W31** and **W32** can be significantly suppressed, and the loss of the EMC filter can also be reduced. Therefore, the performance of the filter structure **300** can be efficiently improved. Moreover, with the first and second openings **S310** and **S320** formed on the bottom side of the first and second magnetic conductive members **S31** and **S32**, the size and height of the filter structure **300** along the **Z** axis can be reduced, thereby saving costs and achieving miniaturization of the product.

[0060] In some embodiments, the second choke element **32** may be omitted from the filter structure **300**, and only the first choke element **31** and the capacitor element **C** are disposed on the substrate **B**. However, the configuration of the filter structure **300** is not limited to those disclosed in the embodiments of the invention.

[0061] FIG. 8 shows a perspective view of a filter structure **400** in accordance with another embodiment of the invention. As shown in FIG. 8, the filter structure **400** is different from the filter structure **300** of FIG. 7 in that the first magnetic conductive element **S41** of the first choke element **41** and the second magnetic conductive element **S42** of the second choke element **42** both have a C-shaped structure.

[0062] Here, the first magnetic conductive element **S41** forms a first opening **S410** facing the $-X$ direction, and the second magnetic conductive member **S42** forms a second opening **S420** facing the **X** direction. That is, the first opening **S410** and the second opening **S420** face opposite directions.

[0063] In this embodiment, the first choke element **41**, the capacitor element **C** and the second choke element **42** are sequentially arranged along the central axis **A4** of the first and second through holes **E410** and **E420**. In this configuration, the annular first ferrite core **E41** is partially received in the C-shaped first magnetic conductive member **S41**, and a part of the first ferrite core **E41** protrudes from the first opening **S410** of the first magnetic conductive member **S41** in the $-X$ direction. Similarly, the annular second ferrite core **E42** is partially received in the C-shaped second magnetic conductive member **S42**, and a part of the second ferrite core **E42** protrudes from the second opening **S420** of the second magnetic conductive member **S42** in the **X** direction.

[0064] Moreover, it can be seen in FIG. 8 that the first magnetic conductive member **S41** has a first section **S411**, a second section **S412** and a third section **S413** connected to each other. The first section **S411** of the first magnetic conductive member **S41** is located between the first ferrite core **E41** and the capacitive element **C** along the **X** axis. The second section **S412** of the first magnetic conductive member **S41** is bent relative to the first section **S411** and located on the top side of the first ferrite core **E41**. The third section **S413** of the first magnetic conductive member **S41** is bent relative to the first section **S411**, affixed to the substrate **B**, and situated on the bottom side of the first ferrite core **E41**. As the first and second magnetic conductive members **S41** and **S42** have symmetrical structures, the second magnetic conductive member **S42** will not be described again here for simplification of the disclosure.

[0065] In this embodiment, the capacitive element **C** is located between the first magnetic conductive member **S41** and the second magnetic conductive member **S42** along the **X** axis, and a first slot **B1** and a second slot **B2** are formed on the substrate **B**. During assembly, the first magnetic conductive member **S41** can be inserted through the first slot **B1** on the substrate **B**, and the second magnetic conductive member **S42** can be inserted through the second slot **B2** on the

substrate B, whereby the first magnetic conductive member **S41** and the second magnetic conductive member **S42** are firmly affixed to the substrate B.

[0066] With the first magnetic conductive member **S41** and the second magnetic conductive member **S42** respectively disposed in the first choke element **41** and the second choke element **42**, coupling noise effect between the capacitor element C and the first and the second windings **W41** and **W42** can be significantly suppressed. Therefore, the loss of the EMC filter can be reduced, and the performance of the filter structure **400** can be efficiently improved. Moreover, with the first opening **S410** formed on the left side of the first magnetic conductive members **S41** and the second opening **S420** formed on the right side of the second magnetic conductive member **S42**, the size and height of the filter structure **400** along the X axis can be reduced, thereby saving costs and achieving miniaturization of the product.

[0067] In some embodiments, the second choke element **42** may be omitted from the filter structure **400**, and only the first choke element **41** and the capacitor element C are disposed on the substrate B. However, the configuration of the filter structure **400** is not limited to those disclosed in the embodiments of the invention.

[0068] Although some embodiments of the present disclosure and their advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the disclosure as defined by the appended claims. For example, it will be readily understood by those skilled in the art that many of the features, functions, processes, and materials described herein may be varied while remaining within the scope of the present disclosure. Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, compositions of matter, means, methods and steps described in the specification.

[0069] As one of ordinary skill in the art will readily appreciate from the disclosure of the present disclosure, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps. Moreover, the scope of the appended claims should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements.

[0070] While the invention has been described by way of example and in terms of preferred embodiment, it should be understood that the invention is not limited thereto. On the contrary, it is intended to cover various modifications and similar arrangements (as would be apparent to those skilled in the art). Therefore, the scope of the appended claims should be accorded the broadest interpretation to encompass all such modifications and similar arrangements.

Claims

1. A filter structure, comprising: a substrate; a capacitor element, disposed on the substrate; and a first choke element, disposed on the substrate, wherein the first choke element comprises a first ferrite core, two first windings, and a first magnetic conductive member, and the first windings are wound around the first ferrite core, wherein the first magnetic conductive member surrounds the first ferrite core and is located adjacent to the capacitor element.

2. The filter structure as claimed in claim 1, further comprising a second choke element disposed on the substrate, wherein the capacitor element is situated between the first and second choke elements, the second choke element has a second ferrite core, two second windings and a second magnetic conductive member, and the second windings are wound around the second ferrite core, wherein the second magnetic conductive member surrounds the second ferrite core and is located adjacent to the capacitor element.

3. The filter structure as claimed in claim 1, wherein a part of the first magnetic conductive member is located between the first windings.
4. The filter structure as claimed in claim 1, wherein the substrate forms a plurality of first slots, and the first magnetic conductive member extends through the first slots.
5. The filter structure as claimed in claim 1, wherein the first magnetic conductive member has a rectangular hollow structure.
6. A filter structure, comprising: a substrate; a capacitor element, disposed on the substrate; and a first choke element, disposed on the substrate, wherein the first choke element comprises a first ferrite core, two first windings and a first magnetic conductive member, the first windings are wound around the first ferrite core, and the first ferrite core is situated in the first magnetic conductive member, wherein the first magnetic conductive member is located adjacent to the capacitor element and forms a first opening.
7. The filter structure as claimed in claim 6, wherein the first magnetic conductive member has a U-shaped structure, and the first opening and the substrate are located on opposite sides of the first ferrite core.
8. The filter structure as claimed in claim 6, wherein the first magnetic conductive member has an inverted U-shaped structure, and the first opening faces the substrate.
9. The filter structure as claimed in claim 6, further comprising a second choke element disposed on the substrate, wherein the capacitor element is situated between the first and second choke elements, and the second choke element has a second ferrite core, two second windings and a second magnetic conductive member, wherein the second windings are wound around the second ferrite core, the second ferrite core is situated in the second magnetic conductive member, and the second magnetic conductive member is located adjacent to the capacitor element and forms a second opening.
10. The filter structure as claimed in claim 9, wherein the first magnetic conductive member and the second magnetic conductive member both have a C-shaped structure, and the first and second openings face opposite directions.
11. A filter structure, comprising: a substrate; a capacitor element, disposed on the substrate; and a first choke element, disposed on the substrate and located adjacent to the capacitor element, wherein the first choke element comprises an annular first ferrite core, two first windings and a first magnetic conductive member, the first ferrite core forms a first through hole, the first windings are wound around the first ferrite core, and the first magnetic conductive member is disposed on the first ferrite core, wherein at least a part of the first through hole is covered by the first magnetic conductive member.
12. The filter structure as claimed in claim 11, wherein the projections of the capacitive element and the first ferrite core on a plane perpendicular to the central axis of the first through hole at least partially overlap.
13. The filter structure as claimed in claim 11, wherein the first magnetic conductive member contacts the first ferrite core, and a gap is formed between the first ferrite core and each of the first windings.
14. The filter structure as claimed in claim 11, wherein the first magnetic conductive member has a first section situated between the first ferrite core and the capacitive element.
15. The filter structure as claimed in claim 14, wherein the first magnetic conductive member further has a second section connected to and bent relative to the first section.
16. The filter structure as claimed in claim 15, wherein the first magnetic conductive member further has a third section disposed on the substrate, and the first section is connected between the second and third sections.
17. The filter structure as claimed in claim 11, further comprising a second choke element disposed on the substrate, wherein the second choke element has an annular second ferrite core, two second windings and a second magnetic conductive member, the second ferrite core forms a second

through hole, the second windings are wound around the second ferrite core, and the second magnetic conductive member is disposed on the second ferrite core, wherein at least a part of the second through hole is covered by the second magnetic conductive member.

18. The filter structure as claimed in claim 17, wherein the first choke element, the capacitor element, and the second choke element are arranged along the central axis of the first through hole.
